

# Erik Hill

## AI Evaluation & Testing Engineer

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### SUMMARY

Self-taught AI evaluation and testing engineer building deterministic, replayable systems for measuring coding agents and LLM applications.

Career-changing from professional kitchens, I built a **verifiable-evaluation suite** whose claims are checked rather than asserted: **eleven published evaluation claims are replayed from pinned artifacts in CI**, deterministic, no LLM judges, failures published by design. Also **shipped a game to Google Play**; **30+ public repos**, **two PyPI packages**, **one npm package**.

### SKILLS

**Agentic / AI:** Multi-agent orchestration · LangGraph · Model Context Protocol (MCP) · LLM application development · Tool-calling & guardrails · RAG · Evals & regression gating (nDCG) · Prompt & context engineering · Safety (OWASP LLM Top 10, human-in-the-loop) · Differential-oracle, property-based & mutation testing · No-LLM-judge grading · Pre-registration & minimum-detectable-effect analysis

**Backend & data:** Python · FastAPI · Pydantic · SQLAlchemy 2.0 · PostgreSQL · pgvector · Alembic · REST / OpenAPI

**Frontend / mobile:** TypeScript · JavaScript · React · Next.js 15 · Java · Capacitor · Android (adb) · WebAssembly (Pyodide, TeaVM)

**Ship & verify:** pytest · Vitest · JUnit · jqwik · Playwright · PyPI / npm packaging · GitHub Actions (matrix CI) · Docker · Cloudflare Pages (Wrangler) · Neon

### SELECTED PROJECTS, all public at [github.com/egnar09](https://github.com/egnar09)

- **agent-certlab:** Two real coding agents certified, every verdict regradeable from artifacts. A certification lab built on three **seeded-defect task families** with provable answer keys (incl. **coordinated two-file defects** no single-file patch can pass), a **precondition prover** (clean passes, every seed fails, it refused one of my own defects), and **calibration agents** the harness must separate before certifying.
- **reference-fleet:** An answer key for eval suites: six models each broken **one documented way at an exactly reproducible seeded rate**, so detection rates are measurable against ground truth, a naive suite detects **1 of 6** (paired protocol). Plus a **LoRA-trained native defect**: mixture 0.507 → realized **0.20 greedy**, **0.31 at temp 1.0**: training's rate control is lossy, measured.
- **vac-protocol:** **Verifiable Agent Claims**: an evidence-bundle format + **offline verifier**; a registry where **eleven real claims are hash-checked and re-earned in CI**, with a **public record of it rejecting my own evidence**. Plus **vac-gate**: replayable evidence or no green check.
- **evalmut:** **Mutation testing for eval graders (PyPI, with its gradecore engine)**: inject known defects, report which checks stayed green. Found **4 holes in its own engine's suite**, **8 in ports of a widely-used framework's assertions**. **No LLM judge**; every operator tied to a documented real failure. *319 tests*.
- **rag-eval-lab:** deterministic RAG **eval harness** that catches a planted hallucination in CI; from-scratch **BM25 matching the published SciFact baseline (0.664 vs 0.665)**. *45 tests*.
- **agent-graph:** a **LangGraph ReAct agent** whose multi-step tool use is **unit-testable**: a swappable mock/LLM policy so the graph runs deterministically without a key, a **safe AST calculator** instead of eval, and a max-step guard. *31 tests*.
- **model-drift:** public board, **frozen hash-fingerprinted suite** across **17 models / 5 labs**, with a derived **minimum-detectable-regression floor**.
- **Also public:** **crashkit** (live) · **mcp-tools** (MCP server, no SDK) · **llm-wire-stub** (npm) · **two open PRs to Promptfoo**.

### REVIEWED BY OTHERS

- **External audit of vac-protocol:** an outside engineer falsified my own verifier, cheapest forgery **four bytes**. Fix merged, then I found the same defect class **four more times** (arXiv, below).
- **Peer review with Bartłomiej Nawara, PhD**, senior AI/ML engineer: his missing-not-at-random critique of my rate-limit handling shaped **model-drift's reliability floor**, credited in the repo. I found **two design faults** in his pre-registered study, both adopted.
- **Merged upstream fix to TeaVM's Java class library**, reviewed and merged by its maintainer.

### PUBLICATIONS

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“Four Ways to Forge a Bundle My Own Verifier Calls Clean: Refusal-Site Mutation Testing of an Evidence-Bundle Verifier.” Erik Hill. **arXiv preprint, 2026**. [arXiv:2608.26183](https://arxiv.org/abs/2608.26183)  
**21 articles** on LLM evaluation failures, **88+ comments** from **26+ readers**. [dev.to/agentdev9](https://dev.to/agentdev9) · **Drift Notes:** [erikhill.dev/notes](https://erikhill.dev/notes)

## EXPERIENCE

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**Independent AI Evaluation & Testing Engineer. Solo Practice** · *Remote* · *Mar 2026 – Present*

- Designed a five-role agent harness: a **four-stage production loop** (Strategy → Execution → Critic → Ops) that plans, builds, reviews and validates changes to a real React / Capacitor product, human-in-the-loop on every consequential step, plus an **asynchronous Evaluation harness** that scores completed batches.
- Built per-role model routing and hook-based automation that hands work between stages with no manual relaunch; enforced rules with **property-based invariants** (jqwik) over a native engine and **adb-driven on-device validation**. Write-up: [github.com/egnar09/agent-dev-harness](https://github.com/egnar09/agent-dev-harness)

**Indie Game Developer. SeraphLight Studios** · *Remote* · *Nov 2025 – Present*

- **Shipped Tap Dodge Rush to Google Play** (Kotlin / Java, Android Views + Canvas, AdMob), closed testing with 12+ testers, then store review, then launch. Building a second, larger game (React / Capacitor), the product the harness above operates on.

**Kitchen Manager / Shift Lead. Big Island Brewhaus & other high-volume kitchens** · *Hawaii* · *2018 – 2025*

- Led shifts and trained staff; owned prep, inventory, and food-safety under sustained pressure.

## EDUCATION

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**Clark College**, 2007–2009 (some college, no degree). · **freeCodeCamp**: self-directed software curriculum.